

Supplementary Information: Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

This Supplementary Information collects the detailed design-, fabrication-, and test-development records supporting the main manuscript. Unlike the main text, the SI references the project’s development history directly, including specific pull requests (PRs) in the project repository (<https://github.com/verticalcloud-lab/tensegrity-optimization>).

S1 Joint-Design Study

The strut-to-cable junction was developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants). The working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies; a captive TPU core is routed inside the PLA shell so that the soft tendon is protected from layer-line failure at the transition.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in [PR #39](#) (joint design) and [PR #35](#) (CAD parameterization and print history).

Placeholder: joint-design variants and the selected dovetail/anchor-bulb junction (import figures from PR #39 / PR #35).

Figure S1: Five candidate strut-to-cable joint geometries and the captive-TPU-core dovetail junction selected for the working prototype.

S2 Support-Generation Workflow for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. The slicer profile combines a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside, producing TPU-safe coverage of vertical members without slicer-side painting.

The manual narrowing-pillar workflow and the threshold-angle recipe are documented in [PR #66](#); the H2D multi-part material assignment fix is in [PR #64](#); the single-batch Sobol plate-layout generator is in [PR #65](#).

Placeholder: manual narrowing-pillar supports vs. default slicer supports for a vertical TPU tendon (import figures from PR #65 / PR #66).

Figure S2: Manual narrowing-pillar support generation for near-vertical TPU tendons.

S3 Drop-Tower Characterization Notes

The primary drop fixture is a bungee-assisted laboratory drop tower whose base accelerates downward faster than $1g$; unconstrained specimens lift off the base during descent by design of the rig, so the protocol constrains the specimen top through light tethers and registers the base. Protocol details, the accelerometer-tuning campaign, and the impact-window analysis are recorded in [PR #67](#), [issue #71](#), and [PR #74](#).

S4 Metal-Analog Validation Plan

A final validation campaign rebuilds a subset of the T_3 -prism designs as metal analogs—hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members—using the same R , H , θ , and member-diameter parameterization as the printed PLA/TPU specimens. To probe transfer across material systems rather than only confirming the optimum, the selected designs deliberately span the predicted performance range: representatives the surrogate predicts to be worst-, mid-, and best-performing are each built in both material systems. Agreement between the measured PLA/TPU ranking and the hollow-aluminum/stainless-steel ranking (reported in the main-text metal-analog validation table and figure) indicates that the optimization generalizes beyond the as-printed polymers.

Placeholder: assembled metal-analog specimens (hollow aluminum rods + stainless-steel threaded cables) beside their printed PLA/TPU equivalents for the worst/mid/best predicted tiers (photographs pending fabrication).

Figure S3: Metal-analog validation specimens alongside their printed PLA/TPU equivalents for each predicted performance tier.